Generation Models Know Space: Unleashing Implicit 3D Priors for Scene Understanding

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Research Questions

- Can implicit 3D priors learned by video generation models be repurposed to improve downstream 3D scene understanding in MLLMs?
- How can generative features from video diffusion models be effectively integrated with semantic features to overcome spatial blindness in multimodal models?
- What is the relationship between multi-view consistency in video generation models and their effectiveness for 3D spatial reasoning tasks?

Core Idea

The paper addresses the spatial blindness problem in MLLMs - while they excel at semantic understanding, they struggle with geometric reasoning. The key insight is that video generation models must implicitly learn robust 3D priors to synthesize temporally coherent videos with consistent occlusion, depth, and motion. Rather than training new 3D modules or using explicit 3D supervision, VEGA-3D extracts these implicit geometric priors from intermediate layers of pre-trained video diffusion models during controlled noise injection. These generative features are then fused with semantic features via an adaptive gating mechanism, providing spatial anchors that enhance localization and geometric understanding without requiring 3D annotations.

Key Formula

Forward Noising

$$z_k = (1 - t_k) z_0 + t_k \epsilon$$

z_k is the noised latent at timestep k, z_0 is clean latent, t_k=k/K is normalized time, epsilon is Gaussian noise. This perturbs clean video latents to activate the generative model's 3D reasoning capabilities during denoising.

Adaptive Gate

$$g_i = \sigma(W_g^\top \operatorname{Concat}(\operatorname{LN}(F_{\text{gen},i}), \operatorname{LN}(F_{\text{sem},i})) + b_g)$$

g_i is scalar gate for token i, W_g is learnable weight, F_gen and F_sem are generative and semantic features, LN is layer norm. Computes token-level weights to balance semantic vs geometric cues for each spatial location.

Key Figure

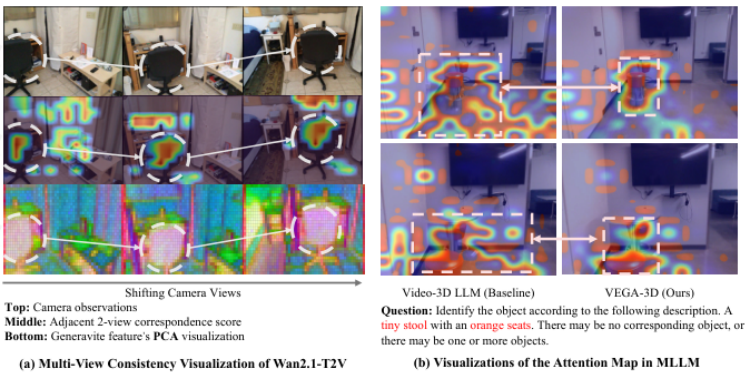

Fig. 2: Visualization of implicit 3D priors. (a) The generation model demonstrates strong multi-view geometric consistency, evidenced by high correspondence scores and stable PCA feature representations across shifting camera views. (b) By leveraging these priors, our **VEGA-3D** overcomes the spatial ambiguity observed in the baseline, yielding precisely-located attention maps of the target object in the instruction.

Vmem [44]) as a *Latent World Simulator* to enrich the visual stream with spatiotemporal world-knowledge priors, complementary to the semantic encoder. To solve the distribution shift between generative and semantic space, we design a token-level adaptive gated fusion module that integrates the two features. This fusion enables the model to actively exploit the generative backbone's 3D awareness to strengthen geometry-sensitive reasoning, while preserving discriminative semantic cues.

Extensive experiments on 3D scene understanding (e.g., visual grounding, dense captioning, and QA), spatial reasoning benchmarks (e.g., VSI-Bench [73]), and robotics manipulation tasks (LIBERO [48]) demonstrate that our method significantly outperforms larger spatially-enhanced models. Furthermore, in Fig. 3, we provide quantitative evidence for the strong correlation between multi-view correspondence and downstream understanding performance. Besides, as evidenced in Fig. 3(a), the gains stem from synergy rather than replacement: generative and semantic features are complementary, and their fusion yields substantial improvements. Our analysis further shows that the most informative spatial cues emerge from intermediate representations and mid-denoising time of the generative model, instead of the final pixel outputs, and that these priors are particularly beneficial for localization-centric tasks, effectively providing a spatial anchor for MLLMs.

In summary, our contributions are threefold.

- We investigate that modern video generators learn transferable spatiotemporal priors that encode geometry-consistent structure and motion, and we show that these priors are most informative in intermediate representations and mid-denoising stages.

Figure 2: Demonstrates the core insight showing video diffusion models achieve high multi-view consistency and that incorporating generative priors sharpens scattered attention maps for precise spatial localization.

Limitations

- Incorporates video diffusion backbone which increases inference cost substantially compared to baseline models
- Performance gains are primarily on localization-centric tasks, with slight drops on dense captioning metrics like CIDEr
- Requires careful tuning of noise levels and layer selection from the generative model for optimal performance
- Framework depends on the quality and scale of pre-trained video generation models, limiting applicability to smaller generative backbones